ELEC 2FH4 - Set 8

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Exercise: Given the surface charge density $\rho_s = 2.0\ \mu\text{C/m}^2$ existing in the region $r = 1.0$ m, $0 < \phi < 2\pi$, $0 < \theta < \pi$ and is zero elsewhere (See Figure 8.2) . Find analytically the energy stored in the region bounded by $2.0\text{ m} < r < 3.0\text{ m}$, $0 < \phi < 2\pi$ and $0 < \theta < \pi$. Write a MATLAB program to verify your answer.

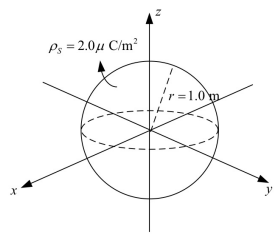

Figure 8.2 The surface charge density $\rho_s = 2.0\ \mu\text{C/m}^2$ at $r = 1.0$ m.

- $\rho_s = 2\mu\ C/m^2$
  - $r = 1m$
  - $0 < \phi < 2\pi$
  - $0 < \theta < \pi$
- $W_\varepsilon = ?$
  - $2 < r < 3$
  - $0 < \phi < 2\pi$
  - $0 < \theta < \pi$

$$Q = \rho_s \cdot A_s$$
$$= \rho_s \cdot 4\pi r^2$$
$$= 2\mu \cdot 4\pi 1^2$$
$$= 8\pi\mu\ C$$

$$\vec{\mathcal{E}}(r) = \frac{Q}{4\pi\varepsilon_0 r^2}\vec{a_r}$$
$$\left|\vec{\mathcal{E}}(r)\right| = \frac{Q}{4\pi\varepsilon_0 r^2}$$

$$u_\varepsilon = \frac{1}{2}\varepsilon_0 \mathcal{E}^2$$

$$W_\varepsilon = \int_v u_\varepsilon\, dV$$
$$= \int_2^3\int_0^\pi\int_0^{2\pi} \frac{1}{2}\varepsilon_0 \mathcal{E}^2\, r^2 \sin\theta\, d\phi\, d\theta\, dr$$
$$= \int_2^3\int_0^\pi\int_0^{2\pi} \frac{1}{2}\varepsilon_0 \frac{Q^2}{(4\pi\varepsilon_0)^2 r^4}\, r^2 \sin\theta\, d\phi\, d\theta\, dr$$
$$= Q^2 \frac{1}{2\cdot 4^2\pi^2\varepsilon_0}\int_2^3\int_0^\pi\int_0^{2\pi} \frac{1}{r^4}\, r^2 \sin\theta\, d\phi\, d\theta\, dr$$
$$= Q^2 \frac{1}{32\pi^2\varepsilon_0}\cdot\int_2^3 \frac{1}{r^2}\, dr \cdot \int_0^\pi \sin\theta\, d\theta \cdot \int_0^{2\pi} d\phi$$
$$= Q^2 \frac{1}{32\pi^2\varepsilon_0}\cdot \frac{-1}{r}\Big|_2^3 \cdot -\cos\theta\Big|_0^\pi \cdot \phi\Big|_0^{2\pi}$$
$$= Q^2 \frac{1}{32\pi^2\varepsilon_0}\cdot \frac{1}{6}\cdot 2\cdot\pi$$
$$= Q^2 \frac{1}{48\pi\varepsilon_0}$$
$$= (8\pi\mu)^2 \frac{\cancel{336\pi}\,M}{\cancel{448\pi}}$$
$$\therefore W_\varepsilon = 0.4737\ J$$